

Hanock Babu Gandu

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Transferable Iqama | Available Immediately

Professional Summary

Full-stack Software Engineer with over 4 years of experience building and scaling healthcare platforms at Astek (KSA) and IQVIA (India). Node.js, AngularJS, React, Express, MongoDB, and healthcare integrations (HL7, Mirth Connect, NPHIES (Clinical & RCM), Sehhaty, Wasfaty, Mawid, Ayenati etc.).

Skills

Frontend : HTML5, CSS3, Bootstrap, i18n, AngularJS, ReactJS, Material-UI, Tailwind.

Backend : Express.js, Node.js, Mongoose, FastAPI, Python, Rest APIs.

AI & Agents : CrewAI, OpenRouter, Hugging Face, Multi Agent Orchestration, LLM Integration, Google Gemini API, LiteLLM, Prompt Engineering.

Databases & Stores : MongoDB, replica sets, mongodump, mongorestore.

Languages : JavaScript, Python, Java.

Cloud & Devops : Git, GitHub, GitLab, Render, Vercel, CI/CD, Docker, Kubernetes.

Tools : Postman, Teleport, PAM, RDP, Robo 3T.

Security : JWT, bcrypt, RBAC, audit logging, captcha, CVE remediation, npm dependency upgrades
Wireshark, tcpdump, Burp Suite.

Debugging : Chrome Dev Tools, React Dev Tools.

Interpersonal : Teamwork, Customer Relations, Cross-functional collaboration.

Work Experience

Full Stack Developer

Aug 2024 - July 2026

ASTEK Saudi Arabia Ltd. (Onsite - Lean Business Services) Riyadh, Saudi Arabia

- Full-stack engineer working for Lean across the Saudi healthcare product suite — **Raqeem** (Ministry of Health), **Rabet+** (Ministry of Defense), and **Shifaa** (Ministry of Interior) — a family of Hospital Information Systems sharing a common core and tailored per ministry. Built citizen-facing Primary Health Center (PHC) and hospital workflows on Node.js / Express.js / AngularJS / ReactJS / MongoDB / Mirth Connect / JasperReports stack, powering **2,000+ PHCs**, **100,000+ active clinical users**, and **200,000+ patient orders per day** across OPD, IPD, Radiology, Lab, Pharmacy, OR, ER, Blood Bank, EMR, Billing, and Revenue-Cycle Management workflows.
- Contributed to integrations across the Saudi healthcare ecosystem, grouped broadly into **identity & national registries** (Yakeen), **citizen-facing apps** (Sehhaty — personal health, Mawid — central appointments), **clinical systems** (Ayenati — lab, NVR — National Vaccine Registry, care team, Seha, TeleHealth - Telemedicine), and **regulatory & e-health platforms** (NPHIES — unified health record, Wasfaty — e-prescription, Sick Leave, Healthy Marriage Screening, School Screening).

Key Initiatives:

- **Smart PHC — Sehhaty integration (flagship initiative):** Built and shipped a real-time patient-callup notification flow that bridges Raqeeem HIS with the Saudi Ministry of Health **Sehhaty** citizen app. Patients making registration, self check-in through Sehhaty app by themselves, and getting notified, instructed by the clinical staff from Raqeeem through push notifications dispatched directly to the patient's Sehhaty app.

- **FMS (Feedback Management System):** Designed and implemented a configuration-driven, **role- and action-aware feedback platform**. Survey popups are triggered by specific events tied to user roles — patient registration for *registration clerks*, medicine dispense for *pharmacists*, result entry for *radiologists*, EMR vitals capture for *nurses*, and other role/action combinations — each with role-specific question sets and an optional free-text feedback field at the end. Administrators configure surveys, target roles, trigger actions, and rollout policy entirely through the application UI without any code changes or redeployments rather just by configuration setting.

Engineering Ownership:

- Worked on end-to-end delivery of multiple Raqem HIS modules - REST API design, Mongoose schema modeling, and business logic - partnering with implementation, QA, and clinical teams.
- Auth & Security: migrated the authentication framework from conventional session-based auth to JWT, hardening token issuance and validation; remediated high- and critical-severity CVEs through targeted package upgrades.
- Production incident triage: troubleshooting and operational support across a **multi-tier deployment** (multiple Node.js application servers under PM2, interface engine(Mirth Connect) servers, MongoDB cluster nodes, and MySQL servers). SSH'd into servers, correlated **application logs (Winston)**, **PM2 process logs**, **Elastic APM transaction traces** to identify root causes and resolve issues.
- API performance engineering: split APIs from monolithic backend service to multiple microservices as **in-memory cached read-only APIs** (alternative endpoints for repetitive, low-logic data-pull calls) that reduced DB round-trips and lowered p95 latency under heavy concurrent load.
- Environment setup (Dev / QA): stood up Dev and QA environments end-to-end - performed **MongoDB mongodump / mongorestore** snapshots from UAT, exported and re-imported **Mirth Connect channels** across Mirth integration servers, configured application and interface server clusters, validated HL7 message replay, supported routine backup, restore for both Mongo and Mirth.
- Background processing (Batch Jobs): built recurring cron jobs with node-schedule for closing patient visits, patient appointments and other business-rule-driven lifecycle events.
- Collaborated with architect engineers, product, and implementation teams on requirement analysis and solution design; performed code reviews, mentored newly onboarded engineers - knowledge transfer on the techno-functional workflows.

Tech stack: Node.js, Express.js, AngularJS, ReactJS, JavaScript, MongoDB, Mongoose, REST APIs, HL7 v2.x, Mirth Connect, JWT, OAuth, JasperReports, Elastic APM, Kibana, Winston, PM2, GitLab CI/CD, Nexus, Linux, Docker, K8s, Postman, mongodump / mongorestore, Mirth channel import/export.

Full Stack Developer (MEAN, MERN)

May 2022 - April 2024

IQVIA Consulting and Information Services India Private Limited - Chennai, India

- Developed comprehensive and highly configurable Hospital Information System (**HIS**) using **MEAN & MERN** Stack technologies, providing hospitals with advanced tools for streamlined clinical task management and patient administration.
- As part of the Product Development team, have worked on user stories and bugs associated with implementing new functionalities for various clients as well as the core product (**HIS/EMR**).
- Began by contributing to the existing frontend technology of the product, AngularJS. Subsequently, transitioned to a new team (R&D) of 4, tasked with developing the frontend of the product from scratch, utilizing **ReactJS** and **Material UI Library (MUI)**.
- Played a pivotal role as a MEAN & MERN Stack Developer in designing and developing the core product component of the IQVIA Hospital Information System, Arcus Air.
- Designed React component architecture and parent-child data flows for static and dynamic data; built CRUD surfaces across multiple pages, consuming REST APIs via Axios.

- Actively participated in requirement analysis and solution discussions; collaborated with 3 other developers in the process of developing the **R&D** core product.

Tech stack: JavaScript, Node.js, Express.js, MongoDB, Mongoose, AngularJS, ReactJS, Material-UI (MUI), HTML5, CSS3, Bootstrap, Git, GitLab, Robo 3T, Chrome DevTools, React DevTools, Postman.

Projects

LinguaTeach - AI-Powered Personal Language Teacher

April 2026

(<https://lingua-teacher.vercel.app/>)

- Built a full-stack web application that acts as a personal language teacher for English/Arabic learning. The app features a conversational AI teacher that explains in English and teaches Arabic with proper script, transliteration, and harakat. Implemented a spaced repetition system for vocabulary retention, text-to-speech with automatic language detection (English and Arabic voices in the same message), browser-based speech recognition for pronunciation feedback, and AI-generated grammar lessons with interactive exercises. Designed a stateless backend with multi-provider AI routing (OpenRouter free models + Gemini 2.5 Flash) for fastest response. Deployed on Vercel (frontend) and Render (backend) at zero cost.

Technologies used: React 19, FastAPI, Python, OpenRouter API, Google Gemini, Edge-TTS, REST API, CI/CD, Vercel, Render.

Courses

- AI Engineering Course (Currently Pursuing)

Education

CVR College of Engineering, Hyderabad, India

2020

Bachelor of Technology (Mechanical Engineering)
